

Decision Trees & Ensemble Methods

Summary Map

1. Decision Trees

When to Use:

- Classification or regression tasks
- Need for interpretability and visualization
- Mixed data types (categorical/numeric)
- Nonlinear relationships

Advantages:

- Easy to interpret and explain
- No need to encode categorical variables
- Handles both regression and classification
- Visual representation (tree diagrams)

Disadvantages:

- Prone to overfitting (especially deep trees)
- High variance, sensitive to data changes
- Not as accurate as ensemble or more complex models

Key Details:

- **Splitting Criteria:** Gini index, Entropy, Classification error
- **Parameters:** max_depth, min_samples_split, min_samples_leaf, criterion
- **Pruning:** Reduces overfitting; use cost-complexity pruning and cross-validation to select optimal tree size
- **Data Types:** Handles both categorical and numeric predictors

Plots & Diagnostics:

- Tree diagrams (structure, splits)
- Feature importance plots
- Confusion matrix (classification)
- Residual plots (regression)
- Learning curves (train/test error vs. tree size)

Metrics for Success:

- Classification: Accuracy, precision, recall, F1-score, confusion matrix
- Regression: MSE, RMSE, MAE, R^2

2. Bagging

When to Use:

- High-variance models (e.g., decision trees)
- To reduce overfitting and improve stability

Advantages:

- Reduces variance, improves accuracy
- Out-of-bag (OOB) error estimation (no need for separate test set)
- Parallelizable

Disadvantages:

- Does not reduce bias
- Less interpretable than a single tree
- Computationally intensive

Key Details:

- **Process:** Train multiple models on bootstrap samples, aggregate predictions (majority vote for classification, mean for regression)
- **Parameters:** n_estimators (number of trees), base estimator parameters
- **OOB Error:** Use unused samples for each tree to estimate error

Plots & Diagnostics:

- OOB error vs. number of trees
- Feature importance (averaged across trees)

Metrics for Success:

- OOB error, accuracy, MSE, etc.

3. Random Forests

When to Use:

- Many features, complex data
- Need for robust, accurate models with minimal tuning

Advantages:

- Reduces variance further by decorrelating trees
- Handles high-dimensional data
- Provides feature importance
- Robust to outliers and noise

Disadvantages:

- Less interpretable than single trees
- Computationally intensive

Key Details:

- **Process:** Like bagging, but at each split, only a random subset of features is considered
- **Parameters:** n_estimators, max_features (mtry), max_depth, min_samples_leaf
 - For classification: $\text{max_features} \approx \sqrt{(\text{number of features})}$
 - For regression: $\text{max_features} \approx \text{number of features} / 3$
- **Feature Importance:** Based on decrease in impurity

Plots & Diagnostics:

- Feature importance bar charts
- Error vs. number of trees
- MSE vs. mtry (number of features per split)

Metrics for Success:

- Accuracy, MSE, OOB error, feature importance

4. Boosting

When to Use:

- Need for high predictive accuracy
- Data is relatively clean (not too noisy)
- Willing to tune more parameters

Advantages:

- Can achieve state-of-the-art accuracy
- Reduces both bias and variance
- Works with various data types

Disadvantages:

- Prone to overfitting if not regularized
- Sequential (not parallelizable)
- More complex, less interpretable

Key Details:

- **Process:** Sequentially fit models to correct previous errors; combine predictions with weights
- **Parameters:** `n_estimators` (number of trees), `learning_rate` (shrinkage, e.g., 0.01–0.1), `max_depth` (usually shallow, e.g., 1–3), `subsample`
- **Types:** AdaBoost, Gradient Boosting, XGBoost, etc.

Plots & Diagnostics:

- Training/test error vs. number of trees
- Error vs. learning rate
- Feature importance

Metrics for Success:

- Accuracy, MSE, AUC, cross-validation error

5. Comparison Table

Method	Overfitting	Interpretability	Main Tuning Parameters	Best For	Key Metric
Decision Tree	High	High	max_depth, min_samples	Simple, interpretable	Accuracy/MSE
Bagging	Low	Medium	n_estimators	Reducing variance	OOB error
Random Forest	Low	Medium	n_estimators, mtry	High-dimensional data	OOB error, MSE
Boosting	Medium	Low	n_estimators, learning_rate, max_depth	High accuracy, low bias	CV error, AUC

6. Implementation Tips

- **Decision Trees:** Use for quick, interpretable models; prune to avoid overfitting.
- **Bagging:** Use when single trees overfit; check OOB error for validation.
- **Random Forests:** Default to random forest for robust, accurate models; tune mtry and n_estimators.
- **Boosting:** Use for maximum accuracy; tune learning rate and tree depth; monitor for overfitting.

References:

- DATA 5322 Lecture Materials
- ISLR2 package documentation
- R package documentation for tree, randomForest, gbm